Highlight stress test: word and sentence rectangles

Every word has a ground-truth rectangle in highlight-stress.groundtruth.json. Each case targets one way a highlight can drift.

A1	p.2	Narrow vs wide glyphs
A2	p.2	Long single-chunk lines
A3	p.2	Monospace control
A4	p.2	Condensed, geometric and old-style fonts
A5	p.2	Extreme sizes in one line
A6	p.2	Drop cap
A7	p.2	Script font with overhangs
B1	p.3	Letter spacing (Tc)
B2	p.3	Justification with word spacing (Tw)
B3	p.3	Manual justification, one Tj per word
B4	p.3	Horizontal scaling (Tz)
B5	p.3	TJ kerning instead of spaces (pdfTeX)
B6	p.3	Per-glyph positioning
B7	p.3	Negative letter spacing
C1	p.4	Symbols from a second font
C2	p.4	Inline math
C3	p.4	Superscript citations and footnotes
C4	p.4	Chemical subscripts
C5	p.4	Bold / italic inside one word
C6	p.4	Size change mid-sentence
C7	p.4	Coloured and serif/sans switch
D1	p.5	Ligature glyphs
D2	p.5	No-break and thin spaces
D3	p.5	Private Use Area in ToUnicode
D4	p.5	Precomposed vs decomposed diacritics
D5	p.5	Hyphenation across lines
D6	p.5	Dashes, quotes, URLs, numbers
D7	p.5	Very long words
D8	p.5	Soft hyphen and zero-width characters
E1	p.6	Hebrew (RTL, visual order)
E2	p.6	Mixed LTR + RTL line
E3	p.6	Arabic presentation forms
E4	p.6	CJK without spaces
E5	p.6	Vertical CJK as stacked glyphs
E6	p.6	Greek and Cyrillic
F1	p.7	Rotated text
F2	p.7	Font size inside the text matrix
F3	p.7	Non-uniform scale and shear
F4	p.7	Text inside a Form XObject
F5	p.7	Mirrored text
G1	p.8	Two columns, 6 pt gutter
G2	p.8	Content stream order ≠ reading order
G3	p.8	Table with aligned numbers
G4	p.8	Fake bold by double printing
G5	p.8	Invisible text
G6	p.8	Microprint and huge display text
H1	p.9	Good OCR layer (per-word)
H2	p.9	Sloppy OCR layer (per-line)
H3	p.9	Skewed scan
H4	p.9	Outlines without any text
P1	p.10	Page with /Rotate 90
P2	p.11	CropBox smaller than MediaBox
P3	p.12	Small page size (A6)
P4	p.13	Landscape MediaBox (no /Rotate)
M1	p.14	Composite two-column article 1
M2	p.15	Composite two-column article 2
N0	p.16	Running header and page number
N1	p.16	Chapter opener on a coloured band
N2	p.16	Three-line drop cap
N3	p.16	Raised initial and small-caps lead-in
N4	p.16	Text wrapping around a photo
N5	p.16	Inline icons inside sentences
N6	p.17	Sentence interrupted by a figure
N7	p.17	Sidebar box (key term)
N8	p.17	Margin note
N9	p.17	Bulleted and numbered lists
N10	p.17	Exercise blanks and checkboxes
N11	p.17	Footnote markers and footnotes
N12	p.18	Pull quote
N13	p.18	Picture initial
N14	p.18	Outlined initial and boxed initial
N15	p.18	Diagram with scattered labels
N16	p.19	Chart with text baked into the image
N17	p.19	White text over a photo
N18	p.19	Photo covering part of the text
N19	p.19	Photo inside a column

Why each case breaks highlights: see README.md and "cases" in the ground-truth JSON.

A. Glyph widths in proportional fonts

A1 narrow vs wide glyphs in one Tj (character-count interpolation fails)

iiiiiiii WWWWWWWW illicit MOMENTUM ill-will mmmmmm lilt WAWA iii MMM

A2 one Tj per full-width line: the error accumulates toward the right edge

Measurements of the shadow of the black hole in M87 imply a mass of six and a half billion solar masses. The ring diameter was measured with very long baseline interferometry at a wavelength of one point three millimetres. Calibration of the station gains removed most of the systematic error, yet the remaining uncertainty grows toward the edge of the image. Independent pipelines produced consistent images, which

A3 monospace control line (every method should be exact here)

```
Monospaced text has equal advances: iiii WWWW mmmm 0123
```

A4 condensed / geometric / old-style faces (text-layer fallback font differs most)

Avenir Next Condensed Bold squeezes every glyph horizontally

Futura Medium has very round geometric letters like OQCG

Baskerville old-style serif with delicate hairlines

A5 36 pt and 5 pt in the same text object, plus a 4.5 pt line

Huge tiny tiny tiny medium

This line is set at four and a half points, which is about the size of a footnote in a dense journal.

A6 drop cap: the word "The" is split into a 40 pt "T" and an 11 pt "he"

The first paragraph starts with a large initial letter that spans three lines of body text, which is common in magazines and books.

A7 calligraphic font: glyph ink overhangs the advance box

Calligraphy overflows

B. Spacing operators: Tc, Tw, Tz, TJ

B1 letter-spaced heading, Tc = 4

L E T T E R S P A C E D H E A D I N G

B2 justified paragraph, one Tj per line, gaps stretched with Tw (Word style)

Measurements of the shadow of the black hole in M87 imply a mass of six and a half billion solar masses. The ring diameter was measured with very long baseline interferometry at a wavelength of one point three millimetres. Calibration of the station gains removed most of the systematic error, yet the remaining uncertainty grows toward the edge of the image. Independent pipelines produced consistent images, which strongly suggests that the ring is not an artefact of the imaging

B3 every word is its own text object placed by x (no space characters at all)

Each word of this sentence is placed separately by the producer

B4 horizontal scaling Tz = 60 and Tz = 150

Horizontally condensed to sixty percent of the width

Stretched to one hundred fifty

B5 LaTeX/pdfTeX style: no space characters, word gaps are TJ numbers, plus intra-word kerning

Words are separated only by kerning offsets inside a TJ array like pdfTeX output

AVATA WAVE Toyota

B6 every glyph is a separate text object with slight jitter

Every glyph here is positioned individually

B7 negative Tc = -0.8 (tight tracking)

Tightly tracked text squeezes its glyphs together

C. Fonts, sizes and baselines mixed inside a line

C1 symbols from a different font inside a sentence (the M87 drift case)

The shadow diameter is 42 ± 3 μas and the mass is $(6.5 \pm 0.7) \times 10^9\ M_\odot$.

C2 inline math with sub- and superscripts (Ts) and large operators

where $\sum_i x_i^2 \le \int_0^\infty f(x)\,dx$ holds for every sample.

C3 superscripts attached to words: E = mc², citations[3–5], footnote marks

Einstein showed E = mc² in 1905 [12], see also earlier work[3–5] and the footnote[*].

C4 chemistry: subscripts lowered inside words

Water is H_2O, carbon dioxide is CO_2 and glucose is $C_6H_{12}O_6$.

C5 style changes inside a word

This is un**believ***able* and so is re**EVALUATE***d text*.

C6 sizes 9 / 22 / 9 in one sentence

Small words then **BIG WORDS** then small again.

C7 coloured inline link in a sans font

Read the full report at example.org/report before the meeting.

D. Unicode and extraction traps

D1 real ligature glyphs (U+FB00–FB04) inside words

The office files were efficiently flagged, affecting five floors of staff.

D2 no-break space, thin space and narrow no-break space

Accuracy 42 ± 3 μas, distance 16 4 Mpc, share 5 % of 10 000 km.

D3 ToUnicode maps the no-break space to U+F0A0 (Private Use Area), like the M87 PDF

Measured 42 ± 3 μas with 7 stations.

D4 Polish diacritics: precomposed (NFC) line, then decomposed (NFD) line

Zażółć gęślą jaźń, źdźbło i łódź.

Zażółć gęślą jaźń, źdźbło i łódź.

D5 hard hyphenation at line end

Reading aloud needs a careful compre-
hension of where each word starts.

D6 punctuation and tokenizer edge cases

state-of-the-art, word—word, don’t, “quoted”, 3.14159, 1,000.5, −40 °C
https://example.org/a/very/long/path?query=1&x=2 and jan.kowalski@example.com

D7 very long words

Donaudampfschiffahrtsgesellschaftskapitän met
pneumonoultramicroscopicsilicovolcanoconiosis patients.

D8 soft hyphen U+00AD and zero-width space U+200B inside words

auto-matic hyphen-ation and zerowidth joins

E. Scripts and text direction

E1 Hebrew stored in visual order (right-to-left)

שלום עולם זהו טקסט בעברית לבדיקה

E2 English sentence with embedded Hebrew words

The word שלום means peace and עולם means world.

E3 Arabic shaped into presentation forms, visual order

مرحبا بالعالم هذا نص عربي للاختبار

E4 Japanese and Chinese without spaces (word list in ground truth)

日本語のテキストには単語の間にスペースがありません。

这是中文文本没有空格

E5 vertical text: one glyph per line, top to bottom

縦書きの文章

VERTICAL

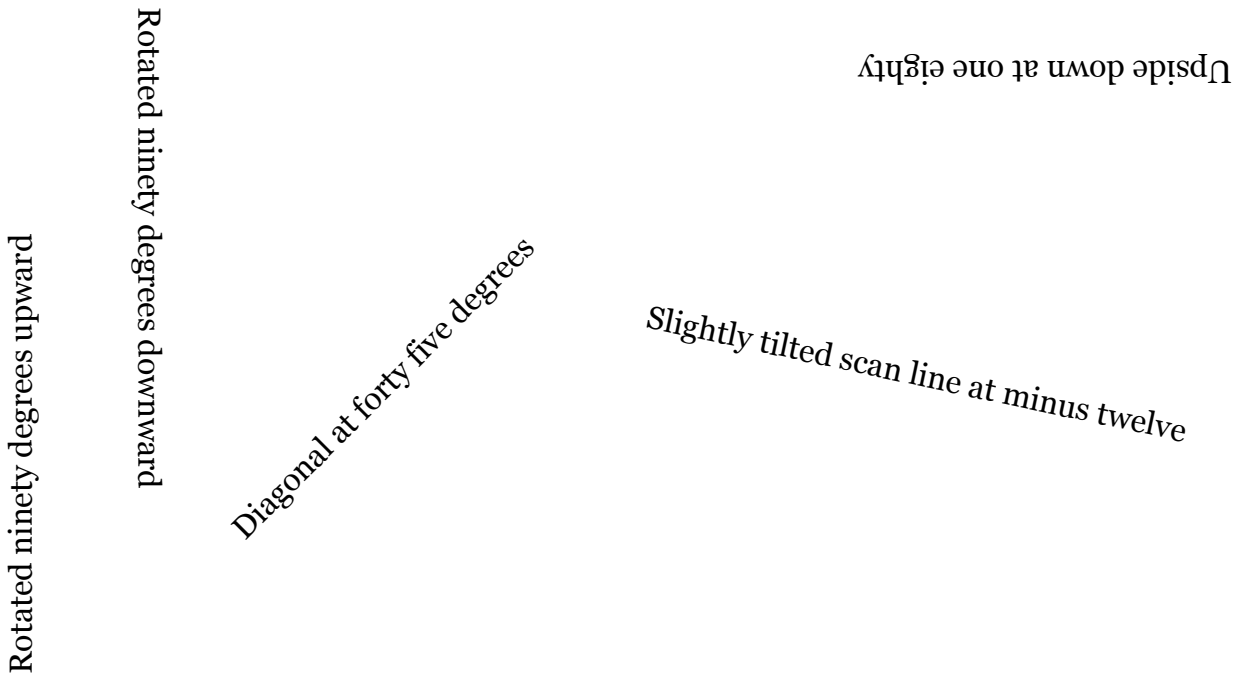
E6 Greek and Cyrillic

Η μέτρηση της σκιάς της μαύρης τρύπας

Измерение тени чёрной дыры в галактике

F. Rotation, matrices and XObjects

F1 text at 90°, -90°, 45°, 180° and -12°



F2 "1 Tf" with the size in Tm (12 0 0 12 x y)

Font size one, text matrix scaled by twelve

F3 non-uniform scale (14 x 9) and synthetic oblique (shear 0.3)

Wide but short letters

Synthetic oblique made by shearing

F4 text inside a Form XObject drawn with scale 0.75 and offset

This paragraph lives inside a reusable form object, which is scaled to three quarters when it is painted.

F5 mirrored text (a = -1), e.g. designs for transparencies

egaмi rorriM

G. Layout, reading order and visibility

G1 two columns separated by only 6 pt

Measurements of the shadow of the black hole in M87 imply a mass of six and a half billion solar masses. The ring diameter was measured with very long baseline interferometry at a wavelength of one point three millimetres. Calibration of the station gains removed most of the systematic error, yet the remaining uncertainty grows toward the edge of the image. Independent pipelines produced consistent images, which strongly suggests that the ring is not an artefact of the imaging algorithm.

G2 same layout, but painted right column first and bottom-up

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G3 table

Station	Baseline km	SNR	Gain error
ALMA	10 800	312.4	1.2%
APEX	9 447	98.1	3.9%
SMT	3 120	57.0	11.0%
IRAM 30 m	9 160	141.7	2.5%

G4 fake bold: identical text painted twice, offset 0.35 pt

Fake bold heading painted twice

G5 hidden text: white text, covered text, invisible render mode (visible=false)

(grey box above hides text)

G6 1 pt microprint and a 60 pt display word

Display

H. Scanned pages with an OCR text layer

H1 glyph outlines (no font) + invisible OCR text fitted per word

Scanned documents carry an invisible text layer
produced by OCR and fitted to every word.

H2 invisible OCR text fitted only per line, shifted 2 pt up

Cheaper OCR engines fit the text per line,
so words inside the line drift noticeably.

H3 page scanned 1.5° crooked, OCR text not rotated

A crooked scan tilts every line slightly,
while the OCR text layer stays perfectly level.

H4 text converted to outlines, no text layer at all (extractable=false)

Outlined text cannot be selected or extracted.

I. Page rotated with /Rotate 90

P1 this page has /Rotate 90; the text below is upright on screen

Measurements of the shadow of the black hole in M87 imply a mass of six and a half billion solar masses. The ring diameter was measured with very long baseline interferometry at a wavelength of one point three millimetres. Calibration of the station gains removed most of the systematic error, yet the remaining uncertainty grows toward the edge of the image. Independent pipelines produced consistent images, which strongly suggests that the ring is not an artefact of the imaging algorithm.

Not rotated in user space, so vertical on screen.

J. CropBox offset and clipped text

P2 CropBox is inset; the words at the page edges are partly or fully cut off

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e left edge of crop

clipped at the r

K. A6 page

P3 a different page size inside the same file

Measurements of the shadow of the black hole in M87 imply a mass of six and a half billion solar masses. The ring diameter was measured with very long baseline interferometry at a wavelength of one point three millimetres. Calibration of the station gains removed most of the systematic error, yet the remaining uncertainty grows toward the edge of the image. Independent pipelines produced consistent images, which strongly suggests that the ring is not an artefact of the imaging algorithm.

L. Landscape page with very long lines

P4 landscape MediaBox, no rotation; each line is one long text item

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M. Composite page 1: realistic two-column article

The shadow diameter is 42 ± 3 μas, consistent with a mass of 6.5×10^{9} solar masses. Our efficient pipeline filters the calibrated visibilities before imaging. Station gains were solved with a self-calibration loop at every scan. The ring brightness is asymmetric, with the southern side about twice as bright. Independent teams produced images with different algorithms and found the same structure. A thin space separates thousands in 10 000 km and the narrow no-break space precedes 5 %. Readers expect the highlight to follow the voice word by word, without lag or drift. Systematic errors were estimated from synthetic data with known ground truth. The posterior distribution of the diameter is narrow and nearly symmetric. Future observations at 345 GHz will sharpen the image by about forty percent. The shadow diameter is 42 ± 3 μas, consistent with a mass of 6.5×10^{9} solar masses. Our efficient pipeline filters the calibrated visibilities before imaging. 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M. Composite page 2: realistic two-column article

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CHAPTER 3

The Water Cycle

How water moves between the sea, the sky and the land

N2 three-line drop cap: "W" (54 pt) + "ater" (10.5 pt) is one word

Water moves constantly between the oceans, the air and the land. Every day the sun lifts about one thousand cubic kilometres of water into the atmosphere, and roughly the same amount falls back as rain or snow. Nothing is lost and nothing is created: the molecules in a glass of tap water may once have fallen on the Tatra mountains, flowed down the Vistula and evaporated from the Baltic Sea. Scientists call this endless journey the hydrological cycle.

N3 raised initial "E" (30 pt) + small-caps lead-in (8 pt) + body (10.5 pt) on one baseline

Evaporation is the first step of the cycle. Heat from the sun turns liquid water into vapour. Warm seas, wet soil and even the leaves of plants release water vapour into the air, where it rises with warm currents until it reaches cooler layers of the atmosphere.

N4 text wraps around a photo; caption under the photo

When moist air rises along a mountain slope it expands, because the pressure around it drops. Expanding air cools, and cool air cannot hold as much water vapour as warm air. Once the air is cold enough, the vapour condenses on tiny particles of dust, salt or smoke and forms the droplets that we see as clouds. This is why the windward side of a mountain range is often wet and green, while the other side lies in a dry rain shadow. In southern Poland the slopes of the Tatra receive more than twice as much precipitation as the lowlands only fifty kilometres to the north, and the difference is visible in the forests, the rivers and even in the shape of the villages.



Figure 3.1 Clouds form over the Tatra mountains when rising air cools.

N5 inline icons; the sun icon replaces the word "sun" (extractable=false)

Press ▶ to start the timer, then ■ when the water boils and write down the ° temperature.

In summer the ☀ heats the Baltic Sea to about twenty degrees.

N6 the sentence "The process … clouds." starts above the photo and ends below it

Clouds are not made of vapour, which is invisible, but of liquid droplets and ice crystals. The process repeats every day as water vapour rises, cools and

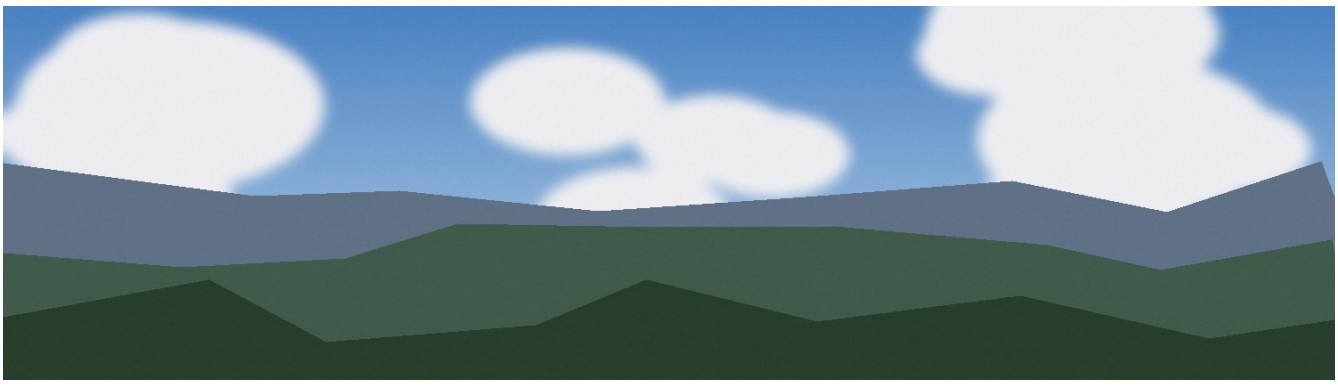


Figure 3.2 Cumulus clouds over the Vistula valley in July.

condenses into tiny droplets that form clouds. A single cumulus cloud can hold more than five hundred tonnes of water.

N7 "Key term" sidebar on the right N8 margin note on the left

Dew forms on surfaces, not in the air. See also p. 47.

At night, grass and car windows are often covered with small drops of water even when it has not rained. The air near the ground cools after sunset, and when it reaches the dew point the vapour condenses directly on cold surfaces. The same process makes a cold glass of lemonade "sweat" on a hot afternoon, and it is the reason why bathroom mirrors fog up after a shower.

KEY TERM

Dew point

The temperature at which air becomes saturated and water vapour starts to condense.

N9 bullets and numbers with hanging indent

- Evaporation: water turns into vapour, mostly over the oceans.
- Condensation: vapour cools and forms droplets around tiny particles in the air, which become clouds and fog.
- Precipitation: rain, snow or hail fall when droplets grow too heavy.

1. Fill a pot with cold water and put a lid on it.
2. Heat it until steam appears, then look at the inside of the lid after two minutes and describe what you see.

N10 exercise with blanks and checkboxes

Exercise 3.4

1. Water boils at _____ °C at sea level.
2. Clouds are made of water vapour. ☐ true ☐ false
3. Name two places where evaporation happens: _____ and _____.

N11 footnote markers in the text and footnotes at the bottom of the page

About 97 percent of the water on Earth is salty[1] and most of the fresh water is frozen in glaciers[2].

[1] Mainly in the oceans; the average salinity is about 35 grams per litre.
[2] Antarctica alone holds about 60 percent of all fresh water.

N12 pull quote

> *Water is the driving force of all nature.*
>
> LEONARDO DA VINCI

N13 ornamental initial "O" is a picture; the text layer starts with "nce"

Once upon a time every drop of water in your glass was part of a cloud, a river, a glacier or the breath of an animal. Water on Earth is very old: most of it arrived more than four billion years ago, and it has been travelling around the planet ever since, changing its form again and again.

N14 "T" drawn as outlines (not text); "R" is white text on a coloured square

The ocean covers seventy one percent of the planet and stores almost all of its water. Its surface layer exchanges heat and moisture with the air every second, which makes the sea the engine of the whole water cycle.

Rivers return the water to the sea. The Vistula carries about one thousand cubic metres every second into the Gulf of Gdańsk, together with sand, mud and dissolved minerals from half of Poland.

N15 vector diagram with labels placed around it

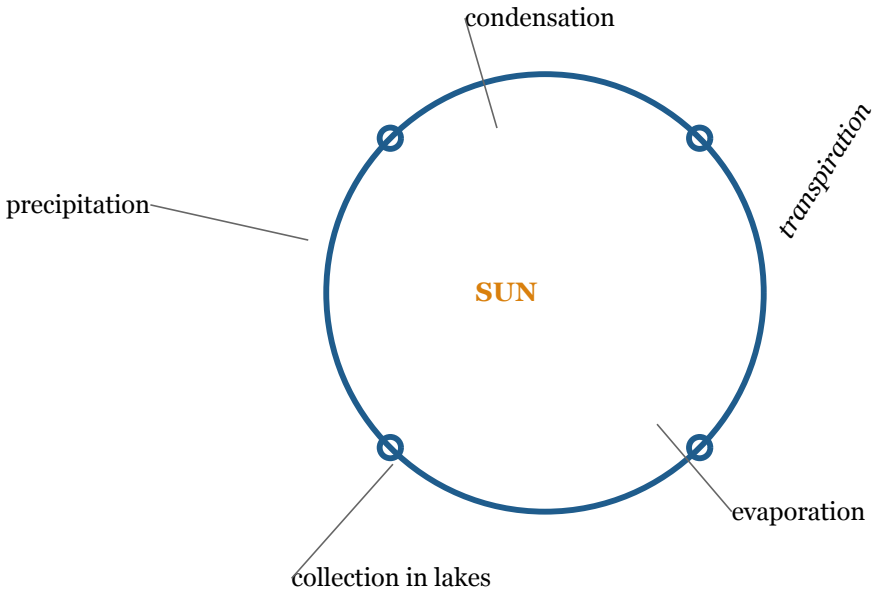
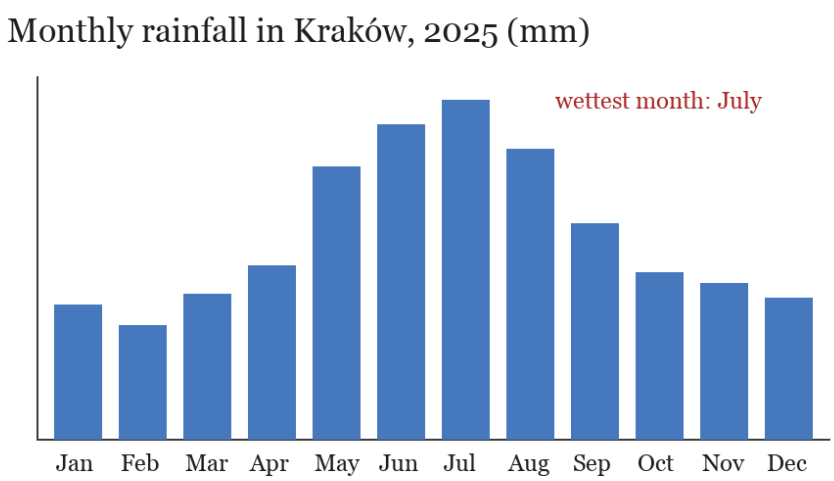


Figure 3.3 The water cycle. Labels are separate text objects around the drawing.

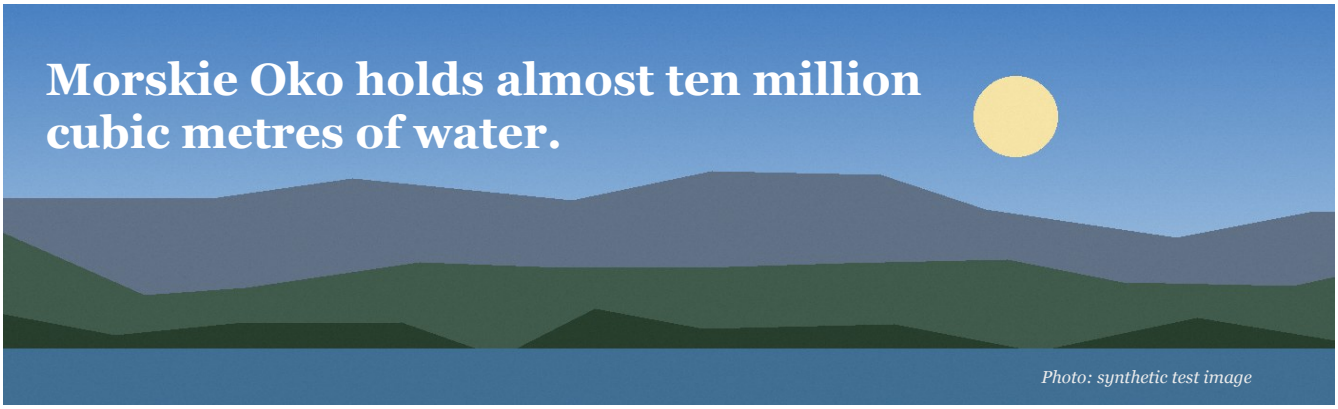
N16 chart image: its title and labels are pixels, not text



The chart shows that the wettest months in Kraków are June, July and August, when short but heavy storms bring most of the yearly rain. A reader that relies only on the PDF text cannot read the chart title or the month names, because they exist only as pixels.

Figure 3.4 Rainfall in Kraków. The title inside the image is not selectable.

N17 white text printed over the photo



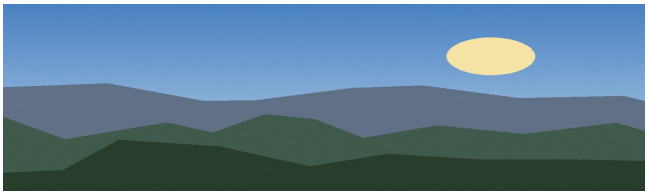
N18 a photo is painted over the middle of this paragraph

Glaciers and snowfields act as slow reservoirs. In winter they store precipitation as ice, and in spring and summer they release it as meltwa driest weeks of the year. When glaciers shrink, rivers first carry more wate



N19 two columns, a photo in the middle of the left column

Rain that falls on a forest does not reach the river at once. Leaves catch part of it, moss and soil soak up the rest.



Only when the soil is full does the water flow over the surface, which is why floods often follow several days of rain.

Groundwater moves much more slowly than rivers. In sand it may travel a few metres a day, in clay only a few centimetres a year. Wells in Poland usually take water from layers between ten and one hundred metres deep, where it has been filtered for decades.